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Republic of the Philippines
CAVITE STATE UNIVERSITY
Bacoor City Campus
SHIV, Molino VI, City of Bacoor

INTERFACES

STEPHEN G. BACOLOR
INSTRUCTOR
DEPARTMENT OF COMPUTER STUDIES

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MISSION

Cavite State University shall provide excellent, equitable and relevant educational opportunities in the arts, science and technology through quality instruction and relevant research and development activities.
It shall produce professional, skilled and morally upright individuals for global competitiveness.

VISION

The premier university in historic Cavite recognized for excellence in the development of globally competitive and morally upright individuals.

STEPHEN G. BACOLOR

Instructor
ITEC 80 – Introduction to HCI
stephen.bacolor@cvsu.edu.ph

COURSE DESCRIPTION

ITEC 80: INTRODCUTION TO HUMAN COMPUTER INTERACTION

The course focuses on the analysis of different user populations with regard to their abilities and characteristics for using both software and hardware products. Evaluation of the design of existing user interfaces based on the cognitive models of the target user is also covered in the course.

PROGRAM OUTCOMES ADDRESSED BY THE COURSE. AFTER COMPLETING THIS COURSE, THE STUDENTS MUST BE ABLE TO:

1. Attain the vision, mission, goals and objectives of the university, campus and department;
2. Deliver a gender fair and gender sensitive instruction to students aligned with university goals and objectives.
3. understand and appreciate the meaning of ethics, values, and attitudes.
4. be guided in their ethical thinking and consideration as they relate in the cyber world; and
5. appreciate and internalize the code of conduct of an I.T. Professional

COURSE REQUIREMENTS

1. Quizzes / Activities / Assignments
2. Class Participation
3. Major Examinations

INTENDED LEARNING OUTCOMES

After the completion of the unit, students will be able to:

1. Understand how interfaces work and its functions
2. Know the different types of user interfaces and understand its implications to different information systems
3. Design an ideal interface for a specific information system

PRE-TEST:

1. What is the main purpose of an interface in computing?
A. To store large amounts of data
B. To connect users and computer systems for interaction
C. To serve as computer memory
D. To secure the operating system

2. Which of the following is an example of a **command-based interface**?
A. Touchscreen ATM
B. Typing commands in a terminal
C. Clicking icons on a desktop
D. Using a VR headset

3. The **WIMP interface** includes all of the following *except*:
A. Windows
B. Icons
C. Menus
D. Programs

4. A **multimedia interface** combines different forms of media such as:
A. Text, audio, video, and images
B. RAM, ROM, and CPU
C. Icons, files, and folders
D. Lights, sensors, and motors

5. Which interface allows users to experience and interact with artificial environments?
A. Command-line interface
B. Virtual Reality
C. Web interface
D. Dashboard interface

6. **Speech interfaces** allow users to interact with systems by:
A. Typing code
B. Speaking and listening
C. Clicking menus
D. Writing text

7. Which interface type provides **tactile feedback** such as vibrations?
A. Haptic
B. Pen-based

C. Air gesture
D. Shareable

8. **Touch interfaces** detect user input by:
A. Typing on a keyboard
B. Clicking a mouse
C. Detecting touch on a display screen
D. Scanning a barcode

9. **Wearable interfaces** are designed to:
A. Be worn by the user and interact with digital systems
B. Replace smartphones
C. Improve operating systems
D. Store files in the cloud

10.Which interface directly connects a user's brain activity to a device?
A. Tangible interface
B. Web interface
C. Brain–Computer Interface
D. Mobile interface

INTERFACES

A. Introduction to Interfaces

Interface and its design process involved working out how best to present information on a screen such that users would be able to:

- perform their tasks
- determining how to structure menus to make options easy to navigate
- designing icons and other graphical elements to be easily recognized and distinguished from one another
- developing logical dialog boxes that are easy to fill in.
- New technologies providing large and small displays

Innovative ways of controlling and interacting with digital information have been developed that include gesture-based, touch-based, and even brain–computer interaction. Researchers and developers have combined the physical and digital interfaces, resulting in various interfaces types.

A major thrust has been to design new interfaces that extend beyond the individual user: supporting small- and large-scale social interactions for people on the move, at home, and at work. There is now a diversity of interfaces, considering the needs in designing interfaces for different environments, people, places, and activities.

B. Interface Types

The interface types are loosely ordered in terms of when they were developed.

1. Command-Based

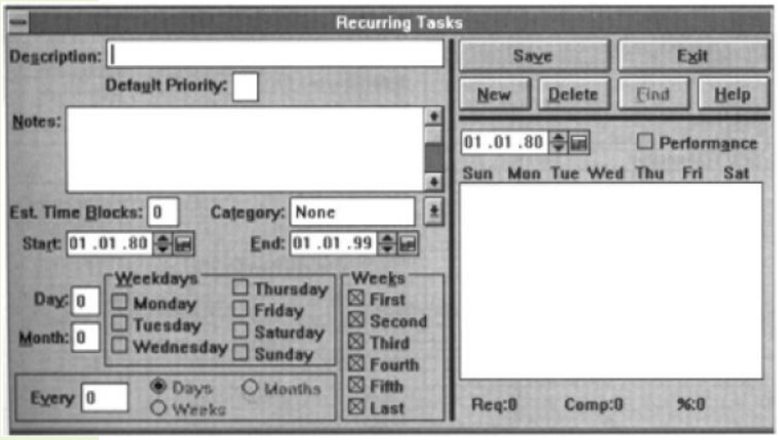
Early interfaces required the user to type in commands that were typically abbreviations at the prompt symbol appearing on the computer display, which the system responded to (e.g. by listing current files using a keyboard).

Another way of issuing commands is through pressing certain combinations of keys (e.g. Shift+Alt+Ctrl). Some commands are also a fixed part of the keyboard, such as delete, enter, and undo, while other function keys can be programmed by the user as specific commands (e.g. F11 standing for print).

Command-Based Interfaces for Virtual Worlds
Commands can be issued to enable the user to move their avatar around, interact with others, and find out about the environment they are in.

The figure shows that the user has issued the command for their avatar to smile and say hello to other avatars sitting by a log fire.

2. WIMP and GUI
This opens up new possibilities for users to interact with a system and for information to be presented and represented at the interface. Specifically, new ways of visually designing the interface became possible, which included the use of color, typography, and imagery. WIMP comprises of:



- **Windows** (that could be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse).
- **Icons** (to represent applications, objects, commands, and tools that were opened or activated when clicked on).
- **Menus** (offering lists of options that could be scrolled through and selected in the way a menu is used in a restaurant).
- **Pointing device** (a mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen).

These include toolbars and docks (a row or column of available applications and icons of other objects such as open files) and rollovers (where text labels appear next to an icon or part of the screen as the mouse is rolled over it).

3. Multimedia
This combines different media within a single interface, namely, graphics, text, video, sound, and animations, and links them with various forms of interactivity. It differs from previous forms of combined media, e.g. TV, in that the different media are interactive. Users can click on hotspots or links in an image or text appearing on one screen that leads them to another part of the program where, say, an animation or a video clip is played. From there they can return to where they were previously or move on to another place.

Many multimedia narratives and games have been developed that are designed to encourage users to explore different parts of the game or story by clicking on different parts of the screen. An assumption is that a combination of media and interactivity can provide better ways of presenting information than can either one alone.

4. Virtual Reality
Virtual reality (VR) uses computer-generated graphical simulations to create "the illusion of participation in a synthetic environment rather than external observation of such an environment". VR is a generic term that refers to the experience of interacting with an artificial environment, which makes it feel virtually real.

Images are displayed stereoscopically to the users – most commonly through shutter glasses – and objects within the field of vision can be interacted with via an input device like a joystick. The 3D graphics can be projected onto CAVE (Cave Automatic Virtual Environment) floor and wall surfaces, desktops, 3D TV, or large shared displays, e.g. IMAX screens.

5. Information Visualization and Dashboards


Information visualizations (infoviz) are computer-generated graphics of complex data that are typically interactive and dynamic. The goal is to amplify human cognition, enabling users to see patterns, trends, and anomalies in the visualization and from this to gain insight. Specific objectives are to enhance discovery, decision-making, and explanation of phenomena. Dashboards have become an increasingly popular form of visualizing information.

- Shows screenshots of data updated over periods of time, intended to be read at a glance.
- They tend not to be interactive; the slices of data are intended to depict the current state of a system or process.

6. Web
In web development, graphical design was viewed as a top priority. A goal was to make web pages distinctive, striking, and pleasurable for the user when they first view them and also to make them readily recognizable on their return.

Users, however, often behave quite differently. They will glance at a new page, scan part of it, and click on the first link that catches their interest or looks like it might lead them to what they want. Much of the content on a web page is not read.

If you look at the design of many websites, you will see that the front page presents a banner at the top, a short promotional video about the company/product/service, arrows to the left or right to indicate where to flick to move through pages and further details appearing beneath the home page that the user can scroll through.

7. Consumer Electronics and Appliances
Consumer electronics and appliances include machines for everyday use in the home, public place, or car (e.g. washing machines, DVD players, vending machines, remotes, photocopiers, printers, and navigation systems) and personal devices (e.g. MP3 player, digital clock, and digital camera).
What they have in common is that most people using them will be trying to get something specific done in a short period of time, such as putting the washing on, watching a

program, buying a ticket, changing the time, or taking a snapshot. They are unlikely to be interested in spending time exploring the interface or spending time looking through a manual to see how to use the appliance.

To design the controls to appear on an LCD screen would enable more information and options to be provided, e.g. only toast one slice, keep the toast warm, automatically pop up when the toast is burning. It would also allow precise timing of the toasting in minutes and seconds.

8. Mobile

Mobile devices have become pervasive, with people increasingly using them in all aspects of their everyday and working lives. They have become business tools to clinch important deals; a remote control for the real world, helping people cope with daily travel delay frustrations; and a relationship appliance to say goodnight to loved ones when away from home.

Smartphones can also be used to download contextual information by scanning barcodes in the physical world. Consumers can instantly download product information by scanning barcodes using their iPhone when walking around a supermarket, including allergens, such as nuts, gluten, and dairy.

9. Speech

A speech or voice user interface is where a person talks with a system that has a spoken language application. It is a specific form of natural language interaction that is based on the interaction type of conversing, where users speak and listen to an interface.

Speech-to-text systems have also become popular, such as Dragon Dictate. Speech technology has also advanced applications that can be used by people with disabilities, including speech recognition word processors, page scanners, web readers, and speech recognition software for operating home control systems, including lights, TV, stereo, and other home appliances.

One of the most popular applications of speech technology is call routing, where companies use an automated speech system to enable users to reach one of their services. Many companies are replacing the frustrating and unwieldy touchtone technology for navigating their services (which was restricted to 10 numbers and the # and * symbols) with the use of caller-led speech.

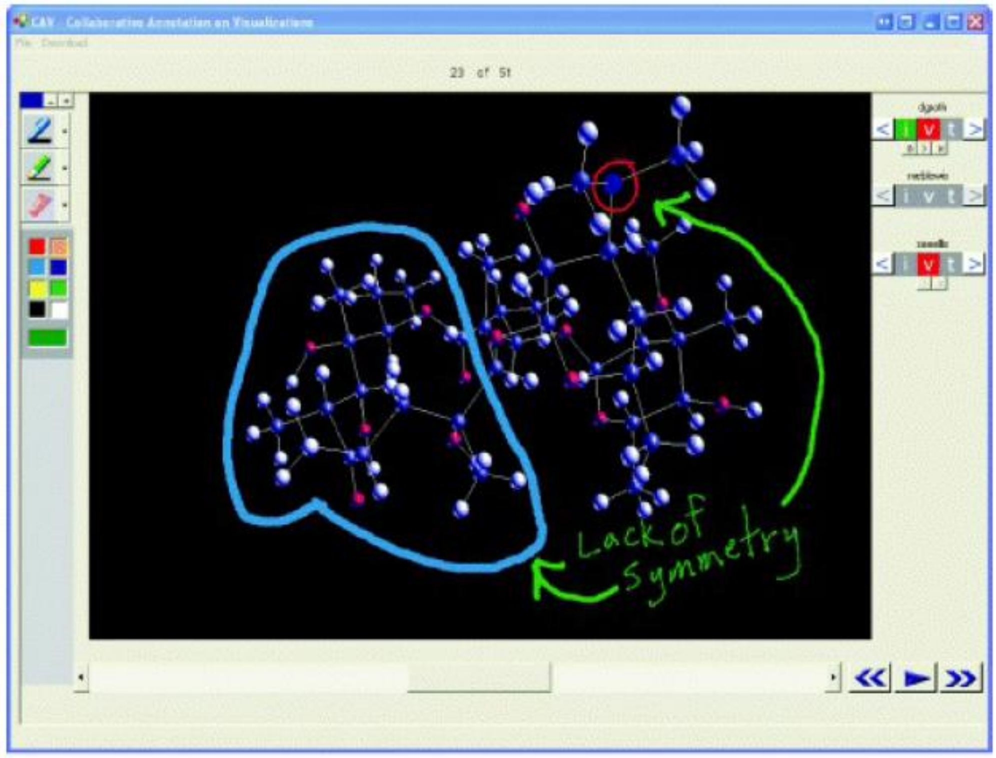
The most common is a directed dialog where the system is in control of the conversation, asking specific questions and requiring specific responses, similar to filling in a form:

System: Which city do you want to fly to?
Caller: London
System: Which airport – Gatwick, Heathrow, Luton, Stansted, or City?
Caller: Gatwick
System: What day do you want to depart?
Caller: Monday week
System: Is that Monday 5th May?
Caller: Yes

10. Pen

Pen-based devices enable people to write, draw, select, and move objects at an interface using light pens or styluses that capitalize on the well-honed drawing and writing skills. They have been used to interact with tablets and large displays, instead of mouse or keyboard input, for selecting items and supporting freehand sketching. Digital ink uses a combination of an ordinary ink pen with a digital camera that digitally records everything written with the pen on special paper.

- The pen works by recognizing a special non-repeating dot pattern that is printed on the paper.
- The non-repeating nature of the pattern means that the pen is able to determine which page is being written on, and where on the page the pen is.
- Infrared light from the pen illuminates the dot pattern, picked up by a tiny sensor.
- The pen decodes the dot pattern as the pen moves across the paper and stores the data temporarily in the pen.
- The digital pen can transfer data that has been stored in the pen via Bluetooth or USB port to a PC.
- Handwritten notes can also be converted and saved as standard typeface text.



11. Touch

Touch screens, such as walk-up kiosks (e.g. ticket machines, museum guides), ATMs, and till machines (e.g. restaurants), have been around for some time. They work by detecting the presence and location of a person's touch on the display; options are selected by tapping on the screen. More recently, multitouch surfaces have been developed as the interface for tabletops and smartphones that support a range of more dynamic fingertip actions, such as swiping, flicking, pinching, pushing, and tapping. These have been mapped onto specific kinds of operations, e.g. zooming in and out of maps, moving photos, selecting letters from a virtual keyboard when writing, and scrolling through lists.

12. Air-Based Gestures

Camera capture, sensor, and computer vision techniques have advanced such that it is now possible to fairly accurately recognize people's body, arm, and hand gestures in a room. An early commercial application that used gesture interaction was Sony's EyeToy, which used a motion-sensitive camera that sat on top of a TV monitor and plugged into the back of a Sony PlayStation.

Sony then introduced a motion-sensing wand, called the Move, that uses the Playstation Eye camera to track players' movements using light recognition technology. Nintendo's Wii gaming console also introduced the Wii Remote (Wiimote) controller as a novel input device. It uses accelerometers for gesture recognition. The sensors enable the player to directly input by waving the controller in front of a display, such as the TV.

The movements are mapped onto a variety of gaming motions, such as swinging, bowling, hitting, and punching. The player is represented on the screen as an avatar that shows him hitting the ball or swinging the bat against the backdrop of a tennis court, bowling alley, or boxing ring. Like Sony's EyeToy, it was designed to appeal to anyone, from young children to grandparents, and from professional gamers to technophobes, to play games such as tennis or golf, together in their living room. The Wiimote also plays sound and has force feedback, allowing the player to experience rumbles that are meant to enhance the experience when playing the game. The Nunchuk controller can also be used in conjunction with the Wiimote to provide further input control. The analog stick can be held in one hand to move an avatar or characters on the screen while the Wiimote is held in the other to perform a specific action, such as throwing a pass in football.

However, sometimes the gesture/body tracking can misinterpret a player's movements, and make the ball or bat move in the wrong direction. This can be disconcerting, especially for expert gamers.

Microsoft's Xbox Kinect comprising an RGB camera for facial recognition plus video capturing, a depth sensor (an infrared projector paired with a monochrome camera) for movement tracking, and downward-facing mics for voice recognition

13. Haptic

Haptic interfaces provide tactile feedback, by applying vibration and forces to the person, using actuators that are embedded in their clothing or a device they are carrying, such as a smartphone or smartwatch. We have already mentioned above how the Wiimote provides rumbles as a form of haptic feedback. Other gaming consoles have also employed vibration to enrich the experience. For example, car steering wheels that are used with driving simulators can vibrate in various ways to provide a feel of the road. As the driver makes a turn, the steering wheel can be programmed to feel like it is resisting – in the way a real steering wheel does.

Providing continuous haptic feedback would be simply too annoying. People would also habituate too quickly to the feedback.

14. Multimodal

Multimodal interfaces are intended to provide enriched and complex user experiences by multiplying the way information is experienced and controlled at the interface through using different modalities, i.e. touch, sight, sound, speech. Interface techniques that have been combined for this purpose include speech and gesture, eye-gaze and gesture, and pen input and speech. An assumption is that multimodal interfaces can support more flexible, efficient, and expressive means of human–computer interaction.

Different input/outputs may be used at the same time, e.g. using voice commands and gestures simultaneously to move through a virtual environment, or alternately using speech commands followed by gesturing. Multimodal systems rely on recognizing aspects of a user's behavior – be it her handwriting, speech, gestures, eye movements, or other body movements.

15. Shareable

Shareable interfaces are designed for more than one person to use. Unlike PCs, laptops, and mobile devices – that are aimed at single users – they typically provide multiple inputs and sometimes allow simultaneous input by collocated groups. These include large wall displays, e.g. SmartBoards, where people use their own pens or gestures, and interactive tabletops, where small groups can interact with information being displayed on the surface using their fingertips. Multiple users can touch the screen at the same time.

An advantage of shareable interfaces is that they provide a large interactional space that can support flexible group working, enabling groups to create content together at the same time.

Compared with a collocated group trying to work around a single-user PC or laptop – where typically one person takes control, making it more difficult for others to take part – large displays have the potential of being interacted with by multiple users, who can point to and touch the information being displayed, while simultaneously viewing the interactions and having the same shared point of reference.

16. Tangible

Tangible interfaces use sensor-based interaction, where physical objects, e.g. bricks, balls, and cubes, are coupled with digital representations. When a person manipulates the physical object(s), it is detected by a computer system via the sensing mechanism embedded in the physical object, causing a digital effect to occur, such as a sound, animation, or vibration. The digital effects can take place in a number of media and places, or they can be embedded in the physical object itself.

Another type of tangible interface is where a physical model, e.g. a puck, a piece of clay, or a model, is superimposed on a digital desktop. Moving one of the physical pieces around the tabletop causes digital events to take place on the tabletop.

The technologies that have been used to create tangibles include RFID tags and sensors embedded in physical objects and digital tabletops that sense the movements of objects and subsequently provide visualizations surrounding the physical objects. Many tangible systems have been built with the aim of encouraging learning, design activities, playfulness, and collaboration.

17. Augmented and Mixed Reality

Augmented reality was mostly experimented with in medicine, where virtual objects, e.g. X-rays and scans, were overlaid on part of a patient's body to aid the physician's understanding of what was being examined or operated on. It was then used to aid controllers and operators in rapid decision-making. One example is air traffic control, where controllers are provided with dynamic information about the aircraft in their section that is overlaid on a video screen showing the real planes landing, taking off, and taxiing.

The additional information enables the controllers to easily identify planes that are difficult to make out – something especially useful in poor weather conditions. Similarly, head-up displays (HUDs) are used in military and civil planes to aid pilots when landing during poor weather conditions. HUD provides electronic directional markers on a fold-down display that appears directly in the field of view of the pilot. Instructions for building or repairing complex equipment, such as photocopiers and car engines, have also been designed to replace paper-based manuals, where drawings are superimposed upon the machinery itself, telling the mechanic what to do and where to do it.

18. Wearables
New flexible display technologies, e-textiles, and physical computing (e.g. Arduino) provide opportunities for thinking about how to embed such technologies on people in the clothes they wear. Jewelry, head-mounted caps, glasses, shoes, and jackets have all been experimented with to provide the user with a means of interacting with digital information while on the move in the physical world. An early motivation was to enable people to carry out tasks (e.g. selecting music) while moving without having to take out and control a handheld device.

Examples include a ski jacket with integrated MP3 player controls that enable wearers to simply touch a button on their arm with their glove to change a track and automatic diaries that keep users up-to-date on what is happening and what they need to do throughout the day.

19. Robots and Drones

The focus has been on designing interfaces that enable users to effectively steer and move a remote robot with the aid of live video and dynamic maps. Domestic robots that help with the cleaning and gardening have become popular. Robots are also being developed to help the elderly and disabled with certain activities, such as picking up objects and cooking meals. Pet robots, in the guise of human companions, are being commercialized. A somewhat controversial idea is that sociable robots should be able to collaborate with humans and socialize with them – as if they were our peers. Several research teams have taken the 'cute and cuddly' approach to designing robots, signaling to humans that the robots are more pet-like than human-like.

Drones are a form of unmanned aircraft that are controlled remotely.

- They were first used by hobbyists and then by the military.
- Entertainment, such as carrying drinks and food to people at festivals and parties
- Agricultural applications, such as flying them over vineyards and fields to collect data that is useful to farmers;
- Helping to track poachers in wildlife parks, and others

20. Brain–Computer Interfaces
Brain–computer interfaces (BCI) provide a communication pathway between a person's brain waves and an external device, such as a cursor on a screen or a tangible puck that moves via airflow). The person is trained to concentrate on the task (e.g. moving the cursor or the puck). Several research projects have investigated how this technique can

be used to assist and augment human cognitive or sensory-motor functions. The way BCIs work is through detecting changes in the neural functioning in the brain.

Every time we think, move, feel, or remember something, these neurons become active. Small electric signals rapidly move from neuron to neuron – that can to a certain extent be detected by electrodes that are placed on a person's scalp. The electrodes are embedded in specialized headsets, hairnets, or caps.

The Brainball game using a brain–computer interface

Brain–computer interfaces have also been developed to control various games. For example, Brainball was developed as a game to be controlled by players' brain waves in which they compete to control a ball's movement across a table by becoming more relaxed and focused.

Other possibilities include controlling a robot and being able to fly a virtual plane by thinking of lifting the mind.

C. Natural User Interfaces and Beyond

Natural User Interfaces and Beyond As we have seen, there are many kinds of interface that can be used to design for user experiences. The staple for many years was the GUI (graphical user interface), which without doubt has been very versatile in supporting all manner of computer-based activities, from sending email to managing process control plants. But is its time up? Will NUIs (short for natural user interfaces) begin to overtake them?

- A NUI is one that enables people to interact with a computer in the same ways they interact with the physical world, through using their voice, hands, and bodies.
- A natural user interface allows users to speak to machines, stroke their surfaces, gesture at them in the air, dance on mats that detect feet movements, smile at them to get a reaction, and so on.
- Refers to the way they exploit the everyday skills we have learned, such as talking, writing, gesturing, walking, and picking up objects.

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Refers to the way they exploit the everyday skills we have learned, such as talking, writing, gesturing, walking, and picking up objects.

Some examples and applications of natural user interfaces:

- Touch screen
- Gesture recognition/ Motion-based

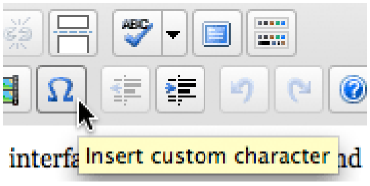
- Speech recognition
- Brain-machine interfaces / Brain-computer interaction

D. Ideal Interface

There is a lot of information out there about various interface design techniques and patterns you can use when crafting your user interfaces and websites, solutions to common problems and general usability recommendations. Following guidelines from experts will likely lead you towards creating a good user interface – but what exactly is a good interface?What are the characteristics of an effective user interface?

Characteristics of an Ideal Interface

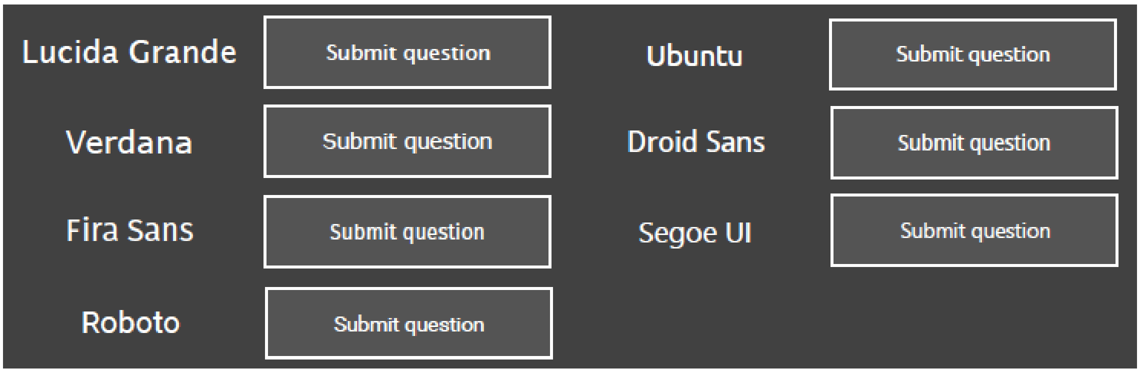
1. Clear



Clarity is the most important element of user interface design. The purpose of user interface design is to enable people to interact with your system by communicating meaning and function. If people can't figure out how your application works or where to go on your website they'll get confused and frustrated.

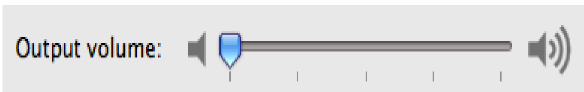
What does that do? Hover over buttons and a tooltip will pop up explaining their functions.

- Only use an icon if its message is a 100% clear to everyone.
- Use clear typography – "All text needs legible typefaces. But especially at interfaces, our eyes need fonts that cooperate rather than resist." —Tobias Frere-Jones
- It has to be clear in small font-sizes



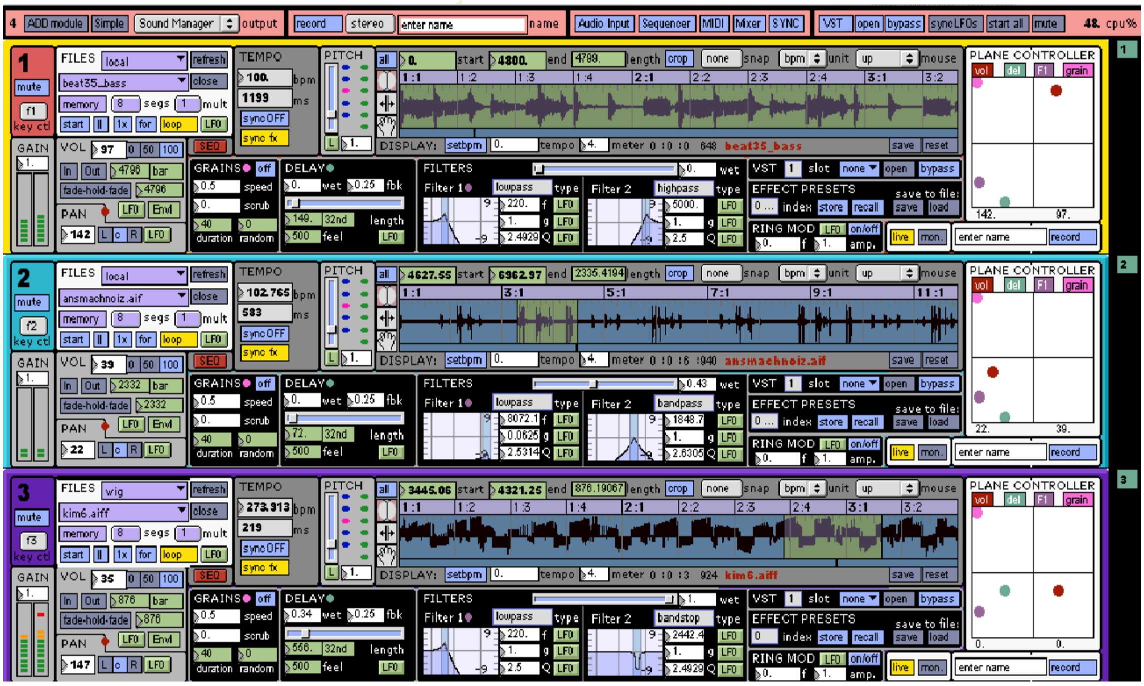
2. Concise

Clarity in a user interface is great, however, you should be careful not to fall into the trap of over-clarifying. It is easy to add definitions and explanations, but every time you do that you add mass. Your interface grows. Add too many explanations and your users will have to spend too much time reading through them.

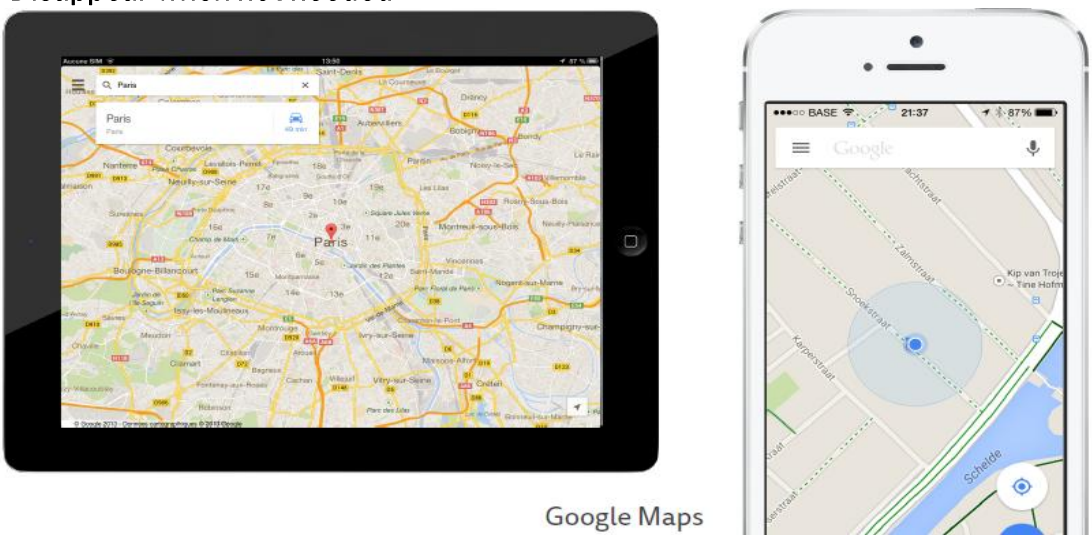


Keep things clear but also keep things concise. When you can explain a feature in one sentence instead of three, do it. When you can label an item with one word instead of two, do it. Save the valuable time of your users by keeping things concise. The volume controls in OS X use little icons to show each side of the scale from low to high.

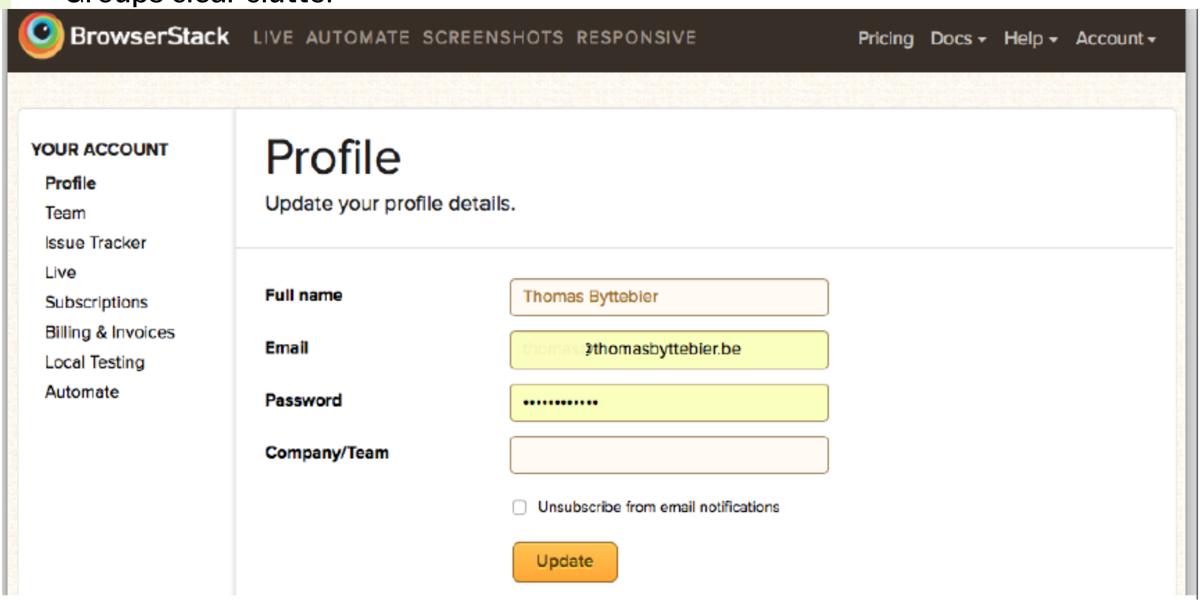
- Every element you add to your UI, makes the rest less important



- Disappear when not needed



- Groups clear clutter



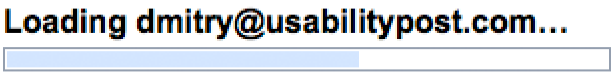
3. Familiar
Many designers strive to make their interfaces 'intuitive'. But what does intuitive really mean? It means something that can be naturally and instinctively understood and comprehended. But how can you make something intuitive? You do it by making it 'familiar'.

Familiar is just that: something which appears like something else you've encountered before. When you're familiar with something, you know how it behaves – you know what to expect. Identify things that are familiar to your users and integrate them into your user interface.

Tabs are familiar because they mimic tabs on folders. You figure out that clicking on a tab will navigate you to that section and that the rest of the tabs will remain there for further navigation.

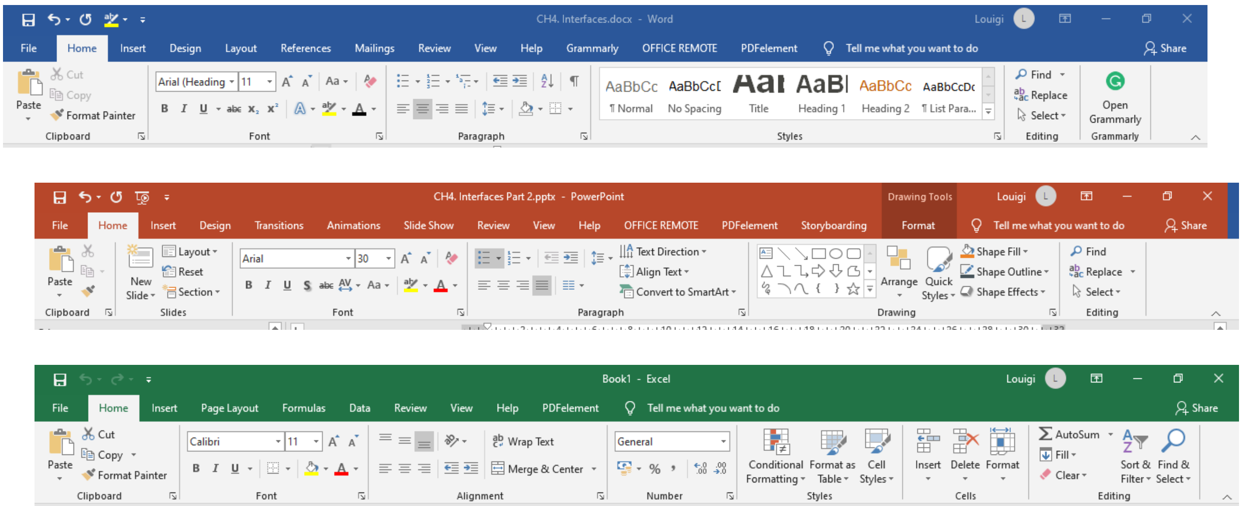
4. Responsive
Responsive means a couple of things. First of all, responsive means fast. The interface, if not the software behind it, should work fast. Waiting for things to load and using laggy and slow interfaces is frustrating. Seeing things load quickly, or at the very least, an interface that loads quickly (even if the content is yet to catch up) improves the user experience.

Responsive also means the interface provides some form of feedback. The interface should talk back to the user to inform them about what's happening. Have you pressed that button successfully? How would you know? The button should display a 'pressed' state to give that feedback. Perhaps the button text could change to "Loading…" and it's state disabled.



Instead of gradually loading the page, Gmail shows a progress bar when you first go to your inbox. This allows for the whole page to be shown instantly once everything is ready.

5. Consistent
Consistent interfaces allow users to develop usage patterns – they'll learn what the different buttons, tabs, icons and other interface elements look like and will recognize them and realize what they do in different contexts. They'll also learn how certain things work, and will be able to work out how to operate new features quicker, extrapolating from those previous experiences.



The Microsoft Office user interface is consistent for a reason.

6. Attractive
This one may be a little controversial but I believe a good interface should be attractive. Attractive in a sense that it makes the use of that interface enjoyable. Yes, you can make your

UI simple, easy to use, efficient and responsive, and it will do its job well – but if you can go that extra step further and make it attractive, then you will make the experience of using that interface truly satisfying. When your software is pleasant to use, your customers or staff will not simply be using it – they'll look forward to using it.

There are of course many different types of software and websites, all produced for different markets and audiences. What looks 'good' for any one particular audience will vary. This means that you should fashion the look and feel of your interface for your audience. Also, aesthetics should be used in moderation and to reinforce function.

- general look & feel
- typography, color, contrast
- visual design trends
- subtle animations and transitions
- subtle affirmative sounds

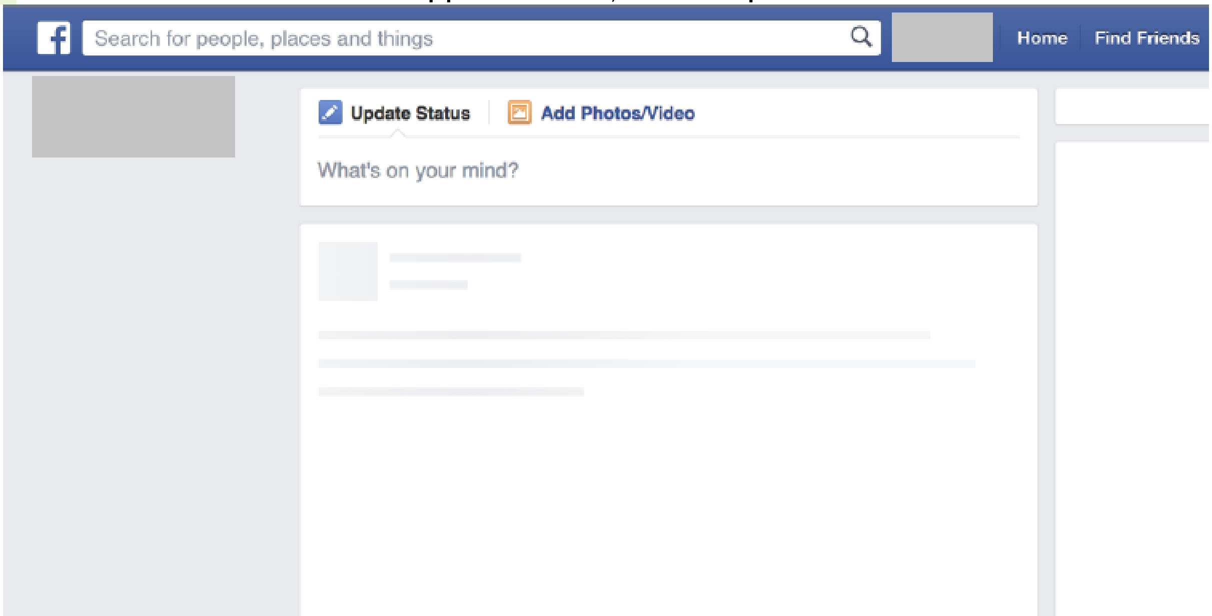
Google are known for their minimalist interfaces that focus on function over form, yet they clearly spent time polishing off the Chrome user interface elements like buttons and icons to make them look just right as evident by the subtle gradients and pixel thin highlights.

7. Fast

Speed is not only a developer's responsibility. As a designer you can create the illusion of speed.

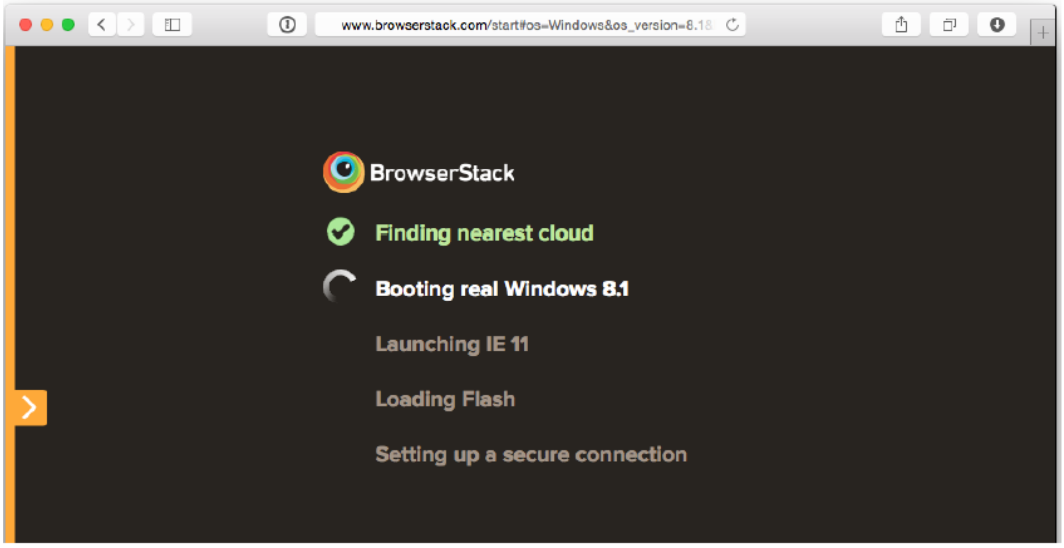
- Show the user something when a page is loading

- A smart UI can make the app feel faster, more responsive



Facebook uses instant placeholders while the real content is loading. Smart trick that makes the app feel faster.

- Also, the user feels more assured



8. Efficient and Effective

A user interface is the vehicle that takes you places. Those places are the different functions of the software application or website. A good interface should allow you to perform those functions faster and with less effort. Now, 'efficient' sounds like a fairly vague attribute – if you combine all of the other things on this list, surely the interface will end up being efficient?

- Figure out what exactly the user is trying to achieve, and then let them do exactly that without any fuss
- Identify how your application should 'work' – what functions does it need to have, what are the goals you're trying to achieve?
- Implement an interface that lets people easily accomplish what they want instead of simply implementing access to a list of features.

Apple has identified three key things people want to do with photos on their iPhone, and provides buttons to accomplish each of them in the photo controls.

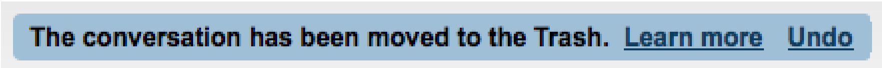
- To design an effective user interface, you must know your user
- Think about novice users vs power users
- Create short cuts that make your app more efficient to those who know where to find them.

9. Forgiving

Nobody is perfect, and people are bound to make mistakes when using your software or website. How well you can handle those mistakes will be an important indicator of your software's quality.

- A forgiving interface is one that can save your users from costly mistakes.
- For example, if someone deletes an important piece of information, can they easily retrieve it or undo this action?

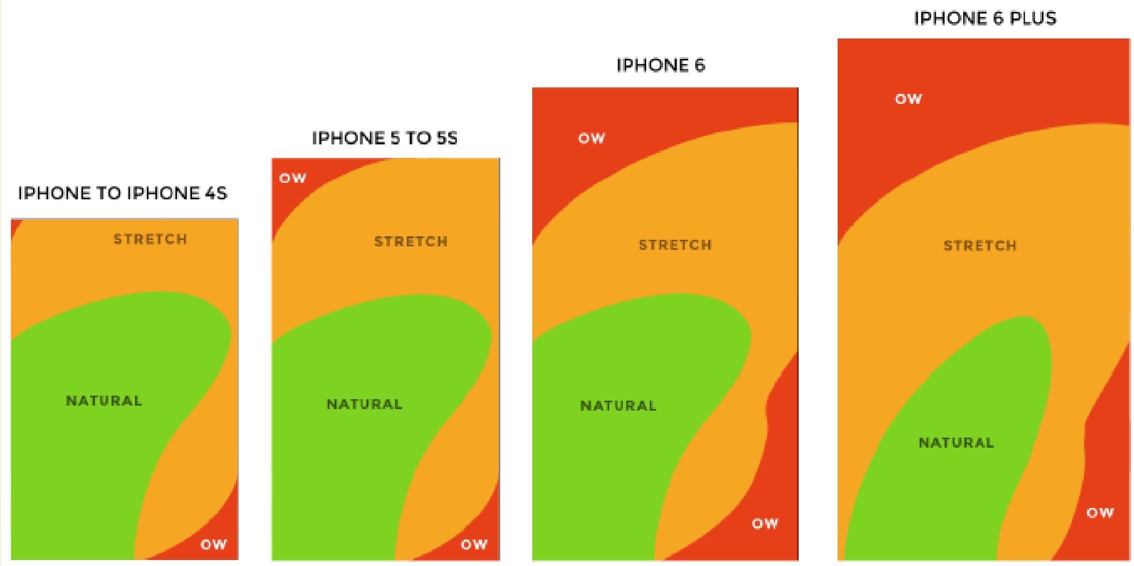
- When someone navigates to a broken or nonexistent page on your website, what do they see?
- Are they greeted with a cryptic error or do they get a helpful list of alternative destinations?



Trashed the wrong email by mistake? Gmail lets you quickly undo your last action.

- Working on achieving some of these characteristics may actually clash with working on others.
    - For example, by trying make an interface clear, you may be adding too many descriptions and explanations, that end up making the whole thing big and bulky.
- Always assume the user will make mistakes while using your interface

10. Make Your Interface More Ergonomic
- place most used buttons in an easy to reach part of the screen
- test as quickly as possible on a real device
- when in doubt, make it larger



With this information, it would be silly to put the most important buttons in the red areas.

POST-TEST:

1. Which of the following best defines an **interface**?
A. A bridge that allows users to communicate with technology
B. A hardware device
C. A storage medium
D. A programming language

2. **Graphical User Interfaces (GUIs)** are based on the concept of:
A. Command lines
B. WIMP — Windows, Icons, Menus, Pointer
C. Dashboards
D. Virtual simulations

3. What is the goal of **information visualization and dashboards**?
A. To amplify human cognition and aid decision-making
B. To provide entertainment
C. To replace text-based input
D. To design games

4. Which interface supports **natural interaction using body movement or gestures**?
A. Command-based interface
B. Air-based gesture interface
C. Web interface
D. Mobile interface

5. The **Haptic interface** enhances user experience by providing:
A. Color feedback
B. Sound feedback
C. Touch or vibration feedback
D. Voice assistance

6. **Tangible interfaces** are unique because they:
A. Use physical objects to manipulate digital information
B. Are limited to text-based input
C. Use only voice commands
D. Work only in VR environments

7. A **Natural User Interface (NUI)** allows people to interact with computers using:
A. Coding syntax
B. Body movements, gestures, and voice
C. Text editors
D. Only mouse clicks

8. Which characteristic of an ideal interface ensures that users understand all functions clearly?
A. Attractive
B. Concise
C. Clear
D. Forgiving

9. An interface that gives users feedback when actions are performed is said to be:
A. Responsive
B. Familiar
C. Consistent
D. Efficient

10. A **forgiving interface** is one that:
A. Prevents users from making any changes
B. Allows recovery from mistakes through undo or error correction
C. Ignores all user errors
D. Requires restarting after an error

ANSWER KEY:

PRE-TEST

MULTIPLE CHOICE

1. B
2. B
3. D
4. A
5. B
6. B
7. A
8. C
9. A
10. C

POST-TEST

MULTIPLE CHOICE

1. A
2. B
3. A
4. B
5. C
6. A
7. B
8. C
9. A
10. B